Conic Sections — Finding and Fixing Sign and Denominator Errors

---

Almost every mistake made with a conic equation is one of two mistakes. Read each equation twice before you compute.

- **Signs.** The standard forms are written with $(x - h)$ and $(y - k)$, so the centre coordinate is the value that makes the binomial *zero*: $(x + 4)$ means $h = -4$, not $h = 4$.
- **Denominators.** Every denominator is a *square*. The semi-axis is its square root, so a denominator of 25 gives a semi-axis of 5; and for a circle the right-hand side is r^2, not r.
- Show the step where you set the binomial to zero and the step where you take the square root. Those are the two steps these errors hide in.

1. Warm-up. Write the centre and the radius of the circle

$$(x + 4)^2 + (y - 7)^2 = 36.$$

Answer: ________________

2. Find and fix. Dev looks at $(x - 5)^2 + (y + 2)^2 = 49$ and writes: “the centre is $(-5,\ 2)$ and the radius is 49.”

Dev made *two* separate errors. Name each one, then write the correct centre and the correct radius.

Answer: ________________

3. Semi-axes of an ellipse. For $\dfrac{x^2}{25} + \dfrac{y^2}{9} = 1$, write the length of the semi-major axis and the length of the semi-minor axis.

Answer: ____________________

4. Back to standard form. The circle $x^2 + y^2 - 6x + 10y + 18 = 0$ is written in general form. Complete the square in x and in y, then write the centre and the radius. Watch the sign of each coefficient and the side of the equation the constant 18 belongs on.

Answer: ____________________

5. Find and fix. Rae studies $\dfrac{x^2}{4} + \dfrac{y^2}{49} = 1$ and reports: "the major axis is horizontal and its length is 4."

Rae used the wrong denominator *and* skipped a step. State which denominator decides the major axis and why, then give the correct direction and the correct full length of the major axis.

Answer: ____________________

6. Shifted ellipse. For $\dfrac{(x+3)^2}{16} + \dfrac{(y-1)^2}{4} = 1$, write the centre and the length of the semi-major axis.

Answer: ____________________

7. Find and fix. Sam writes $\frac{y^2}{9} - \frac{x^2}{16} = 1$ and concludes: "it opens left and right, with vertices 4 units either side of the centre."

Which term carries the plus sign, and what does that tell you about the direction the branches open? Correct both the direction and the vertices.

Answer: __________________

8. Show what you know. Choose one centre other than the origin and use it for all three parts.

(a) Write an equation of a circle with that centre.
(b) Write an equation of an ellipse with that centre whose major axis is vertical.
(c) Write an equation of a hyperbola with that centre whose branches open left and right.
(d) Explain, in two or three sentences, how a reader who sees only your three equations can tell them apart — and how they can read your centre back out of the signs.